Credit Card Fraud Detection

Project Report | Elevate Labs AI/ML Internship

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Abstract

This project implements a comprehensive credit card fraud detection system that identifies fraudulent transactions using a combination of unsupervised anomaly detection and supervised machine learning. The pipeline compares three models -- Isolation Forest, Local Outlier Factor (LOF), and a calibrated XGBoost classifier -- evaluated across ROC-AUC, Precision-Recall, F1-Score, and confusion matrices. A fully interactive Streamlit dashboard provides real-time single-transaction predictions, batch CSV scoring, dynamic threshold control, and SHAP-based explainability. The best model achieves 86.2% F1-Score and 97.7% ROC-AUC on a highly imbalanced dataset of 284,807 transactions with only 0.17% fraud rate.

Introduction

Credit card fraud is a significant challenge for financial institutions, with fraudulent transactions representing a tiny fraction of all activity but causing substantial financial losses. Traditional rule-based systems struggle to keep pace with evolving fraud patterns. This project aims to build a machine learning pipeline that can automatically detect fraudulent transactions in real-time. The Kaggle Credit Card Fraud Detection dataset provides 284,807 transactions over 2 days with only 492 fraud cases (0.17%), presenting a classic highly imbalanced binary classification challenge where standard accuracy metrics are misleading.

Tools Used

- Python 3.10+ - Core development and orchestration
- Pandas & NumPy - Data manipulation and numerical operations
- Scikit-Learn - Preprocessing, anomaly detection, metrics, and calibration
- XGBoost - Gradient-boosted tree classifier with hyperparameter tuning
- imbalanced-learn - SMOTE oversampling inside cross-validation folds
- SHAP - TreeExplainer for model interpretability and feature contributions
- Matplotlib & Seaborn - Static plots: confusion matrices, ROC, PR curves
- Plotly - Interactive charts in the Streamlit dashboard
- Streamlit - Interactive web UI for real-time fraud prediction and batch scoring
- joblib - Model serialization for pipeline, calibrator, and SHAP explainer

Steps Involved in Building the Project

1. Dataset Loading & Exploratory Data Analysis

Loaded the Kaggle Credit Card Fraud Detection dataset containing 284,807 transactions with 30 features (V1-V28 PCA components, Time, Amount) and a binary Class label. EDA revealed extreme class imbalance (578:1 ratio), no missing values, and right-skewed amount distributions. Visualized class distribution with logarithmic scaling and compared transaction amounts between legitimate and fraudulent transactions.

2. Preprocessing & Feature Scaling

Applied RobustScaler (IQR-based, robust to outliers) to Amount and Time columns, which were on very different scales from the PCA-transformed V1-V28 features. Dropped original unscaled columns and reordered the feature set. Performed a stratified 80/20 train/test split (227,845 train / 56,962 test) to preserve the 0.17% fraud ratio in both splits.

3. Unsupervised Anomaly Detection

Trained Isolation Forest (n_estimators=200) and Local Outlier Factor (n_neighbors=20, novelty=True) exclusively on legitimate training transactions -- no fraud labels required. Both models learn what "normal" looks like and flag deviations as potential fraud. Anomaly scores were normalized to [0,1] for consistent evaluation. Isolation Forest achieved 21.1% precision and 27.6% recall; LOF improved to 38.0% precision and 77.6% recall, demonstrating that local density methods suit this dataset better.

4. Supervised XGBoost with SMOTE & Calibration

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Built an XGBoost pipeline with SMOTE (sampling_strategy=0.5) applied only inside cross-validation folds to prevent data leakage. Used RandomizedSearchCV (30 iterations, 5-Fold Stratified CV) optimizing F1-Score. Best params: n_estimators=200, max_depth=6, learning_rate=0.2, subsample=0.9, colsample_bytree=0.7. Applied isotonic calibration on a 15% hold-out split to correct SMOTE-inflated probabilities back to true fraud prevalence (~0.17%). Achieved 90.0% precision, 82.7% recall, 86.2% F1, and 97.7% ROC-AUC.

5. Comprehensive Model Evaluation

Generated ROC curves, Precision-Recall curves, and confusion matrices for all three models. XGBoost dominated across all metrics. Key insight: Precision-Recall AUC (0.8432 for XGBoost) is far more informative than ROC-AUC for imbalanced data. Feature importance analysis identified V14, V10, V4, V17, and V12 as the top fraud indicators, confirmed by both XGBoost F-scores and SHAP contribution analysis.

6. Streamlit Dashboard Deployment

Built a dark-themed interactive Streamlit dashboard with single-transaction input (30 feature fields), batch CSV upload with auto-evaluation against True_Label, dynamic threshold slider (0.01-1.0), real-time classification updates without model re-run, SHAP-based explanations for each prediction, and a calibrated probability gauge. The dashboard loads serialized pipeline, calibrator, and SHAP explainer via joblib.

7. Synthetic Test Data Generation

Created test_fraud.py to generate 40 synthetic transactions (30 legitimate + 10 fraud) using feature distributions derived from the original dataset. Fraud samples have significantly shifted means on key discriminating features (V14 mean ~-6.52, V3 ~-7.50, V7 ~-5.43), enabling instant dashboard validation with known ground truth labels.

Conclusion

The project successfully demonstrates that supervised learning (XGBoost + SMOTE + calibration) significantly outperforms unsupervised anomaly detection for structured credit card fraud data. The calibrated XGBoost model achieves 86.2% F1-Score with only 9 false positives across 56,864 legitimate test transactions, making it production-ready for real-time deployment. Key lessons: (1) SMOTE must be applied inside CV folds only to prevent leakage; (2) isotonic calibration is essential to map SMOTE-trained probabilities back to true prevalence; (3) Precision-Recall metrics are far more informative than accuracy for imbalanced fraud detection. Future work includes real-time API deployment, continuous model retraining with streaming transaction data, and advanced ensemble methods combining multiple supervised models.